

Answer Key & Teacher Notes

Session 1 – Surface Area of Rectangular Prisms · Session 2 – Surface Area of Triangular Prisms

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Surface Area of Rectangular Prisms

TALK PROMPTS

- Why is surface area measured in square units even though the chest is a solid?
- Which two edges make each face? Show your partner on the net.
- Why must you round up when you count cans of stain?

WATCH FOR

- Multiplying $l \times w \times h$ (volume) when surface area is asked.
- Adding the three different faces without doubling them.
- Rounding cans of paint down, or including a face that is not covered.

PARTNER TASK

Gift Wrap. Partners each find the three different face areas of a box 12 in by 10 in by 7 in (120, 84, 70 in²). One partner doubles and adds; the other checks by labeling every face on a net. Both should get 548 in².

SESSION 2 • TEACHER NOTES

Surface Area of Triangular Prisms

TALK PROMPTS

- Why do all three rectangles in this net share the same length?
- How do you decide which measure is the triangle's height and which is just a side?
- Where does the 6 cm show up in each of the cheese wedge's three rectangles?

WATCH FOR

- Using the triangle's height as the side of a rectangle.
- Counting only one triangle end.
- Forgetting the $\frac{1}{2}$ in the triangle area.

PARTNER TASK

Doorstop Check. Partner A finds both triangle ends of a prism 8 cm long with right-triangle ends 6, 8, 10 cm (48 cm²); Partner B finds the three rectangles (192 cm²). Add for 240 cm², then each explains which number was used as a height and which as a side.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.G.A.4	1	2	1	1	1	2	1	3	3	1	1
S1 • Form B	6.G.A.4	1	2	1	1	1	2	1	3	3	1	1
S2 • Form A	6.G.A.4	1	2	2	1	2	2	2	3	3	1	1
S2 • Form B	6.G.A.4	1	2	2	1	2	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

surface area: The total **area** of all the faces of a solid.

congruent: Exactly the same **size** and shape.

net: A flat **pattern** that folds into a solid.

- A rectangular prism has **3** pairs of matching faces.
- $SA = 2lw + 2lh + 2wh$.
- Surface area is measured in **square** units.

Example: Top: $6 \times 4 = 24$ cm². Front: $6 \times 3 = 18$ cm². End: $4 \times 3 = 12$ cm². Double each: $48 + 36 + 24$. Answer: $SA = 108$ cm²

YOU TRY

- $2(5 \times 2) + 2(5 \times 3) + 2(2 \times 3) = 20 + 30 + 12 = 62$ in²
- a. $6 \times 4 = 24$ m² b. $6 \times 100 = 600$ in²

EXIT TICKET

- a. $6 \times 9 = 54$ ft² b. $2(12) + 2(8) + 2(6) = 52$ cm²
- $2(8 \times 5) + 2(8 \times 2) + 2(5 \times 2) = 80 + 32 + 20 = 132$ in²
- B** ($2lw + 2lh + 2wh$)

PRACTICE FORM A

- a. $6 \times 64 = 384$ cm² b. $2(54) + 2(36) + 2(24) = 228$ cm² c. $2(50) + 2(30) + 2(15) = 190$ in² d. $6 \times 36 = 216$ m²
- a. top 15 ft², front 10 ft², end 6 ft² b. $2(15) + 2(10) + 2(6) = 62$ ft² c. $62 \div 20 = 3.1$, so 4 cans (round up)
- Faces 120, 84, 70 in²; $SA = 2(120) + 2(84) + 2(70) = 548$ in²
- a. $6 \times 25 = 150$ cm² b. $6 \times 49 = 294$ cm²
- $2(48) + 2(24) + 2(18) = 180$ ft²
- No. 144 in² is one face. A cube has 6 faces: $6 \times 144 = 864$ in².
- Faces 180, 150, 120 in²; $SA = 360 + 300 + 240 = 900$ in²
- a. 184 ft² b. $184 \div 40 = 4.6$, so 5 cans c. $184 - 40 = 144$ ft²; 3.6, so 4 cans
- Jonah found the volume. Surface area adds the areas of the 6 faces: $2(70) + 2(50) + 2(35) = 310$ cm².
- C** (96 cm²)
- B** (3)

Self-check word SURFACES

PRACTICE FORM B

- a. $6 \times 81 = 486$ cm² b. $2(40) + 2(24) + 2(15) = 158$ cm² c. $2(48) + 2(24) + 2(8) = 160$ in² d. $6 \times 25 = 150$ m²
- a. top 24 ft², front 18 ft², end 12 ft² b. $2(24) + 2(18) + 2(12) = 108$ ft² c. $108 \div 25 = 4.32$, so 5 cans (round up)
- Faces 126, 84, 54 in²; $SA = 2(126) + 2(84) + 2(54) = 528$ in²
- a. $6 \times 36 = 216$ cm² b. $6 \times 64 = 384$ cm²
- $2(63) + 2(36) + 2(28) = 254$ ft²
- No. 121 in² is one face. A cube has 6 faces: $6 \times 121 = 726$ in².
- Faces 160, 128, 80 in²; $SA = 320 + 256 + 160 = 736$ in²
- a. 280 ft² b. $280 \div 50 = 5.6$, so 6 cans c. $280 - 60 = 220$ ft²; 4.4, so 5 cans
- Kara found the volume. Surface area adds the areas of the 6 faces: $2(54) + 2(45) + 2(30) = 258$ cm².
- C** (76 m²)
- B** (square units)

Self-check word PACKAGES

GUIDED NOTES

triangular prism: A solid with 2 triangle **bases** and 3 rectangle faces.

lateral face: A side face that is **not** a base.

surface area: The total area of all the **faces**.

- Add the areas of all **5** faces: 2 triangles and 3 rectangles.
- Each triangle uses $\frac{1}{2} \times \text{base} \times \text{height}$.
- Each rectangle is one triangle **side** $\times$ the prism's length.

Example: One end: $\frac{1}{2} \times 8 \times 3 = 12 \text{ ft}^2$. Both ends: 24 ft^2 . Rectangles: $8 \times 6 + 5 \times 6 + 5 \times 6 = 108 \text{ ft}^2$. Answer: $SA = 24 + 108 = 132 \text{ ft}^2$

YOU TRY

- Ends $2 \times (\frac{1}{2} \times 6 \times 4) = 24 \text{ cm}^2$; rectangles $(6 + 5 + 5) \times 7 = 112 \text{ cm}^2$; $SA = 136 \text{ cm}^2$
- a. $2 \times 10 = 20 \text{ in}^2$ b. 36 in^2

EXIT TICKET

- a. $2 \times 6 = 12 \text{ m}^2$ b. $12 \times 2 = 24 \text{ m}^2$
- Ends $2 \times (\frac{1}{2} \times 10 \times 12) = 120 \text{ in}^2$; rectangles $(10 + 13 + 13) \times 5 = 180 \text{ in}^2$; $SA = 300 \text{ in}^2$
- D** (the triangle's height)

PRACTICE FORM A

- a. $2 \times 12 = 24 \text{ ft}^2$ b. 50 ft^2 c. 18 cm^2 d. $12 \times 11 = 132 \text{ cm}^2$
- a. $2 \times (\frac{1}{2} \times 24 \times 5) = 120 \text{ cm}^2$ b. $24 \times 20 + 13 \times 20 + 13 \times 20 = 1,000 \text{ cm}^2$ c. $120 + 1,000 = 1,120 \text{ cm}^2$
- Ends $2 \times (\frac{1}{2} \times 20 \times 21) = 420 \text{ cm}^2$; rectangles $(20 + 21 + 29) \times 14 = 980 \text{ cm}^2$; $SA = 1,400 \text{ cm}^2$
- Ends 60 cm^2 ; rectangles $30 \times 6 = 180 \text{ cm}^2$; $SA = 240 \text{ cm}^2$
- No. 168 in^2 is one end. Both ends: $2 \times 168 = 336 \text{ in}^2$.
- Ends $2 \times (\frac{1}{2} \times 36 \times 24) = 864 \text{ ft}^2$; rectangles $(36 + 30 + 30) \times 25 = 2,400 \text{ ft}^2$; $SA = 3,264 \text{ ft}^2$
- Ends $2 \times (\frac{1}{2} \times 16 \times 6) = 96 \text{ ft}^2$; rectangles $(16 + 10 + 10) \times 10 = 360 \text{ ft}^2$; $SA = 456 \text{ ft}^2$
- a. $2 \times (\frac{1}{2} \times 8 \times 3) = 24 \text{ ft}^2$ b. $24 + 80 + 2 \times (5 \times 10) = 204 \text{ ft}^2$ c. $204 - 80 = 124 \text{ ft}^2$
- Amara counted only one triangle end. A prism has 2 matching ends: $2 \times 24 = 48 \text{ cm}^2$. $SA = 48 + 240 = 288 \text{ cm}^2$.
- B** (3)
- B** (prism's length)

Self-check word TRIANGLE

PRACTICE FORM B

- a. $2 \times 30 = 60 \text{ ft}^2$ b. 84 ft^2 c. 30 cm^2 d. $24 \times 5 = 120 \text{ cm}^2$
- a. $2 \times (\frac{1}{2} \times 16 \times 6) = 96 \text{ cm}^2$ b. $16 \times 15 + 10 \times 15 + 10 \times 15 = 540 \text{ cm}^2$ c. $96 + 540 = 636 \text{ cm}^2$
- Ends $2 \times (\frac{1}{2} \times 8 \times 15) = 120 \text{ cm}^2$; rectangles $(8 + 15 + 17) \times 10 = 400 \text{ cm}^2$; $SA = 520 \text{ cm}^2$
- Ends 48 cm^2 ; rectangles $24 \times 5 = 120 \text{ cm}^2$; $SA = 168 \text{ cm}^2$
- No. 60 in^2 is one end. Both ends: $2 \times 60 = 120 \text{ in}^2$.
- Ends $2 \times (\frac{1}{2} \times 24 \times 16) = 384 \text{ ft}^2$; rectangles $(24 + 20 + 20) \times 20 = 1,280 \text{ ft}^2$; $SA = 1,664 \text{ ft}^2$
- Ends $2 \times (\frac{1}{2} \times 12 \times 8) = 96 \text{ ft}^2$; rectangles $(12 + 10 + 10) \times 12 = 384 \text{ ft}^2$; $SA = 480 \text{ ft}^2$
- a. $2 \times (\frac{1}{2} \times 6 \times 4) = 24 \text{ ft}^2$ b. $24 + 72 + 2 \times (5 \times 12) = 216 \text{ ft}^2$ c. $216 - 72 = 144 \text{ ft}^2$
- Felix used the triangle's height (4 ft) as a side. The rectangles use the triangle's sides: $6 \times 10 + 5 \times 10 + 5 \times 10 = 160 \text{ ft}^2$. Ends $2 \times (\frac{1}{2} \times 6 \times 4) = 24 \text{ ft}^2$. $SA = 184 \text{ ft}^2$.
- B** (5)
- B** (prism's length)

Self-check word COVERING

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 · SURFACE AREA OF RECTANGULAR PRISMS

1. $2(10 \times 4) + 2(10 \times 5) + 2(4 \times 5) = 80 + 100 + 40 = 220 \text{ cm}^2$
2. a. 54 cm^2 b. 600 ft^2 c. 28 m^2 d. 90 in^2 e. $6 \times 25 = 150 \text{ in}^2$ f. $2(18) + 2(12) + 2(6) = 72 \text{ cm}^2$
3. $SA = 2(96) + 2(48) + 2(32) = 352 \text{ in}^2$. Yes, $400 > 352$, with 48 in^2 to spare.
4. Top $4 \times 2 = 8$; front and back $2 \times (4 \times 3) = 24$; sides $2 \times (2 \times 3) = 12$; $8 + 24 + 12 = 44 \text{ ft}^2$
5. He found the volume (8 cm^3). Surface area adds the 6 faces: $6 \times (2 \times 2) = 24 \text{ cm}^2$.

SHEET 2 · SURFACE AREA OF TRIANGULAR PRISMS

1. Triangles $2 \times (\frac{1}{2} \times 6 \times 4) = 24$; rectangles $6 \times 10 + 5 \times 10 + 5 \times 10 = 160$; $SA = 184 \text{ m}^2$
2. a. $2 \times (\frac{1}{2} \times 8 \times 3) = 24 \text{ ft}^2$ b. 45 ft^2
3. Ends 24 ; floor $8 \times 9 = 72$; sides $2 \times (5 \times 9) = 90$; total 186 ft^2
4. Ends $2 \times (\frac{1}{2} \times 5 \times 12) = 60$; rectangles $5 \times 4 + 12 \times 4 + 13 \times 4 = 120$; $SA = 180 \text{ in}^2$
5. He used the triangle's height (4 m) as a side. The rectangles use the sides of the triangle: 6×10 , 5×10 , and 5×10 .
6. **C** (82 cm^2)
7. **B** (192 cm^2)